3	**(a)** **(i)**	$k = F/x$ **or** 1 / gradient		C1	
		($k = 4.4/(5.4 \times 10^{-2})$ =) 81 (81.48) N m^{-1}		A1	[2]
	(ii)	work done = area under line **or** $\frac{1}{2}Fx$ **or** $\frac{1}{2}kx^2$		C1	
		(= $0.5 \times 4.4 \times 5.4 \times 10^{-2}$ =) 0.12 (0.119) J		A1	[2]
	(b) **(i)**	kinetic energy/E_k <u>of trolley/T</u> (and block) changes to EPE/strain energy/elastic energy <u>of spring</u>		B1	
		EPE changes to KE <u>of trolley/T</u> and KE <u>of block</u> **or** to give <u>lower</u> KE to trolley		B1	[2]
	(ii)	change in momentum	$= m(v + u)$	C1	
			= 0.25 (0.75 + 1.2) = 0.49 (0.488) N s	A1	[2]